

SUMMARY

PhD-trained Astrophysicist with 6 years of research experience applying programming, statistical analysis, and scientific computing techniques to complex observational and simulation datasets. Experienced in Python, Fortran, data processing, visualisation, and high-performance computing environments. Currently transitioning into data science, with a strong foundation in quantitative problem solving, data analysis, and developing reproducible analytical workflows. Seeking Data Scientist or Data Analyst opportunities where strong analytical and technical skills can be applied to real-world business challenges.

TECHNICAL SKILLS

Languages **Proficient:** Python, IDL, Fortran, Bash, Git, LaTeX
 Familiar with: R, SQL, HTML, Power BI

EXPERIENCE

Postdoctoral Research Associate 08/2025 - 11/2025
University of Lancashire Preston, UK

- Developed and maintained computational models in Python and Fortran to analyse complex datasets and investigate data-driven research questions
- Designed and implemented data processing and analysis workflows for large-scale observational and simulation datasets exceeding 20 GB, ensuring accuracy, reproducibility, and efficiency
- Refactored existing HPC pipeline code to be environment-agnostic, and developed Python and Bash automation scripts to streamline job submission and monitoring, reducing manual overhead and improving portability across computing systems.
- Communicated technical findings through reports, presentations, and peer-reviewed publications for both technical and non-technical audiences

Associate Lecturer in Mathematics 10/2024 - 05/2025
University of Lancashire Preston, UK

- Delivered undergraduate mathematics teaching and tutorials to diverse student cohorts
- Communicated complex technical concepts clearly to audiences with varying levels of technical expertise
- Provided mentoring, feedback, and academic support to improve student outcomes

PROJECTS

Training Data Warehouse | SQL, Python, Pandas 2026
Personal Project | <https://github.com/CJInman/Stength-Training-Warehouse>

- Developed relational schema with primary and foreign key constraints across sessions, exercises, and set-level data
- Build ETL pipeline using Pandas to load and validate structured data into MySQL
- Wrote SQL queries using joins, aggregations, and window functions to analyse performance trends and training volume

Training Analytics Pipeline | Python, Pandas, Matplotlib 2026
Personal Project | <https://github.com/CJInman/Training-Analytics>

- Built an end-to-end Python analytics pipeline to transform raw training data into structured performance insights on strength progression and workload trends.
- Processed 2000+ training sessions across 3 core lifts, generating time-series analysis of 40,000+ logged sets and 2 primary performance metrics (estimated 1RM and training volume).
- Visualised strength trends and training load over time
- Used synthetic data to ensure reproducibility and enable public deployment

Personal Project | <https://github.com/CJInman/Personal-Finance-Tracker>

- Built an end-to-end personal finance analytics dashboard in Power BI using transaction data in Excel and Power Query
- Cleaned and transformed transaction data using Pandas and a keyword based merchant classification system.
- Created DAX measures (7-day and 31-day moving averages) to analyse spending trends and increase readability of data.
- Identified key insights in spending behaviour including seasonal cashflow patterns and top expense categories.
- Identified 40% of all expenses were non-essential and produced targeted recommendations to reduce outgoings.
- Produced anonymised synthetic dataset for public portfolio use.

EDUCATION

PhD in Astrophysics

University of Lancashire

Graduated 11/2025

Preston, UK

MSc in Astrophysics

University of Central Lancashire

Graduated 11/2021

Preston, UK

BSc (Hons) Mathematical Physics: 1st Class (85%)

University of Central Lancashire

Graduated 05/2020

Preston, UK

- Best BSc (Hons) on Physics courses (School Prize)